

can find improved algorithms for artefact rejection, too. EEGLAB is a good starting point for those who are trying to master independent component analysis.

The toolkit provides a full structure for description of EEG data. This structure includes signal, trials, channel location, reaction time, type of trials and others. An expert programmer can make use of this structure either from MATLAB command line or in MATLAB scripts.

Research programmers and methods developers. They can share their newly-developed methods with the research community around the world as EEGLAB plugins, which automatically appear in EEGLAB menu and which users can download.

How efficient is EEGLAB in comparison with commercial EEG software

When it comes to making a decision of whether to move along with an open source solution like EEGLAB or rely on expensive proprietary ones, several parameters come into consideration. If you are willing to pay heavily for commercial software, you can expect expert support personnel willing to help you round the clock for your project.

For an open source project like EEGLAB there are no dedicated personnel for user support and the assistance delivered depends on availability and advancement on other projects.

Available features. Although some commercial toolboxes come with more methods for issues like source localisation, EEGLAB offers

Some useful plugins

Given below is the list of some major plugins available with EEGLAB:

FASTER. This is a fully-automated, unsupervised method for high-density EEG data processing.

ERPLAB. This toolbox helps in analysing event-related potential (ERP) data. The plugin extends the capabilities of EEGLAB, providing a robust tool for ERP processing, visualisation and analysis.

NBT. Neurophysiological Biomarker Toolbox (NBT) proposes a standardised way of data storage and analysis so as to allow various research projects to merge, compare or share their data and biomarker algorithms.

NFT. Neuroelectromagnetic Forward Modeling Toolbox (NFT) generates realistic head models from available data and computes numerical solutions for solving the forward problem of electromagnetic source imaging.

SIFT. Analysis and visualisation of multivariate causality and information flow between sources of electrophysiological (EEG/ECOG/MEG) activity can be carried out with the help of this toolbox plugin.

DIPFIT. This toolbox plugin sources localisation of ICA component sources of EEG data.

FMRI. This plugin helps in the removal of functional Magnetic Resonance Imaging (fMRI) artefacts in EEG data.

BCILAB. This plugin toolbox helps to build and test brain computer interfacing models.

MPT. This toolbox is used for comparison and inference of probabilistic multi-subject EEG source data.

far more features than its commercial counterparts. As development is carried out by open source scientific research community, latest developments in the area are often first reflected in EEGLAB and later adopted by commercial software.

Scripting capabilities. EEGLAB makes use of flexibility and the range offered by MATLAB for its scripting. Commercial providers usually rely on their own proprietary language. Running MATLAB code within their scripts limits its capabilities.

Stability of code. Validation of the code is not always strictly carried out as per industry standards. Also, the software is always under constant development. Thus, in comparison with the most-reliable commercial counterpart, stability of the code cannot be guaranteed. If

some stability issues are brought to the notice of developers, these are addressed in the future revisions of the software.

Graphical interface. The graphical user interface of EEGLAB may not be as refined as that of proprietary software. Processing is mainly done in EEGLAB using command line and the graphical user interface is used as just a tool to facilitate processing.

Within three years of its first stable release in 2003, there were around 25,000 downloads for the software from about 73 countries. Today, the toolbox is counted among the most-widely-used signal-processing environments by cognitive neuroscientists for EEG data processing. **EFY**

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